

Worksheet — Motion in a Straight Line (Class 12 NIOS Physics)

Total marks: **30** · Questions: **10**

Instructions: Attempt all 10 questions. Show every step in the subjective questions; choose one option for MCQs. Total time: 50 minutes.

Q1. The SI unit of acceleration is: [1 mark]

- (A) m/s
- (B) m/s^2
- (C) m·s
- (D) $\text{m} \cdot \text{s}^2$

Q2. Distance is always greater than or equal to the magnitude of displacement. [1 mark]

- (A) True
- (B) False

Q3. The third kinematic equation for uniform acceleration is $v^2 = u^2 + \underline{\hspace{2cm}}$. [1 mark]

Q4. A car starts from rest with acceleration 3 m/s^2 . Distance covered in 4 s: [2 marks]

- (A) 6 m
- (B) 12 m
- (C) 24 m
- (D) 48 m

Q5. A ball is dropped from rest and reaches the ground with velocity 20 m/s ($g = 10 \text{ m/s}^2$). Height of drop: [2 marks]

- (A) 10 m
- (B) 20 m
- (C) 40 m
- (D) 80 m

Q6. A car decelerates uniformly from 25 m/s to rest in 10 s. Find the deceleration and the stopping distance. [3 marks]

Q7. A stone is thrown upward at 25 m/s. Take $g = 10 \text{ m/s}^2$. Find (i) maximum height, (ii) time of flight, (iii) velocity 1.5 s after launch. [3 marks]

Q8. A v - t graph is straight from (0, 0) to (8, 16) m/s, then horizontal to (12, 16). Compute (i) acceleration in the first 8 s, (ii) total displacement after 12 s. [4 marks]

Q9. (a) Derive the equation $s = ut + \frac{1}{2} a t^2$ from the definitions of velocity and acceleration. (b) A body moves with $u = 5 \text{ m/s}$, $a = 2 \text{ m/s}^2$. Compute s at $t = 0, 1, 2, 3, 4 \text{ s}$ and tabulate. [5 marks]

Q10. (a) Sketch the v - t graph of a body that is dropped from rest, takes 3 s to fall, hits the ground and bounces back with no energy loss, then rises for 3 s before being caught at the original release height (take downward as positive, $g = 10 \text{ m/s}^2$). (b) Find the total distance and total displacement covered. [8 marks]

Answer Key

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Q1. The SI unit of acceleration is:

Answer: m/s²

$$a = \Delta v / \Delta t \rightarrow \text{m/s} \div \text{s} \rightarrow \text{m/s}^2.$$

MCQ · EASY · 12-NIOS-PHY-MSL-03

Q2. Distance is always greater than or equal to the magnitude of displacement.

Answer: True

Path length $\geq$ straight-line distance between start and end.

TRUE_FALSE · EASY · 12-NIOS-PHY-MSL-01

Q3. The third kinematic equation for uniform acceleration is $v^2 = u^2 + \underline{\hspace{2cm}}$.

Answer: 2as

Eliminates time between $v = u + at$ and $s = ut + \frac{1}{2}at^2$.

FILL_BLANK · EASY · 12-NIOS-PHY-MSL-04

Q4. A car starts from rest with acceleration 3 m/s². Distance covered in 4 s:

Answer: 24 m

$$s = \frac{1}{2} a t^2 = \frac{1}{2} \times 3 \times 16 = 24 \text{ m.}$$

MCQ · EASY · 12-NIOS-PHY-MSL-04

Q5. A ball is dropped from rest and reaches the ground with velocity 20 m/s ($g = 10 \text{ m/s}^2$). Height of drop:

Answer: 20 m

$$v^2 = 2 g h \rightarrow 400 = 20 h \rightarrow h = 20 \text{ m.}$$

MCQ · MEDIUM · 12-NIOS-PHY-MSL-06

Q6. A car decelerates uniformly from 25 m/s to rest in 10 s. Find the deceleration and the stopping distance.

Answer: Deceleration: $a = (0 - 25)/10 = -2.5 \text{ m/s}^2$, so $|a| = 2.5 \text{ m/s}^2$. Stopping distance: $s = (u + v)/2 \times t = (25 + 0)/2 \times 10 = 125 \text{ m.}$ (Or $s = ut + \frac{1}{2}at^2 = 250 - 125 = 125 \text{ m.}$)

Apply $v = u + at$ and the average-velocity formula.

SHORT_ANSWER · MEDIUM · 12-NIOS-PHY-MSL-04

Q7. A stone is thrown upward at 25 m/s. Take $g = 10 \text{ m/s}^2$. Find (i) maximum height, (ii) time of flight, (iii) velocity 1.5 s after launch.

Answer: (i) $h_{\text{max}} = u^2/(2g) = 625/20 = 31.25 \text{ m}$. (ii) $t_{\text{total}} = 2u/g = 50/10 = 5 \text{ s}$. (iii) $v = u - gt = 25 - 15 = 10 \text{ m/s}$, still upward.

Direct application of free-fall formulas.

SHORT_ANSWER · MEDIUM · 12-NIOS-PHY-MSL-06

Q8. A v - t graph is straight from (0, 0) to (8, 16) m/s, then horizontal to (12, 16). Compute (i) acceleration in the first 8 s, (ii) total displacement after 12 s.

Answer: (i) $a = (16 - 0)/(8 - 0) = 2 \text{ m/s}^2$. (ii) Total displacement = area under graph = triangle ($\frac{1}{2} \times 8 \times 16 = 64$) + rectangle ($4 \times 16 = 64$) = 128 m.

Slope = a ; area under v - t = displacement.

SHORT_ANSWER · MEDIUM · 12-NIOS-PHY-MSL-07

Q9. (a) Derive the equation $s = ut + (1/2)at^2$ from the definitions of velocity and acceleration. (b) A body moves with $u = 5 \text{ m/s}$, $a = 2 \text{ m/s}^2$. Compute s at $t = 0, 1, 2, 3, 4 \text{ s}$ and tabulate.

Answer: (a) Average velocity = $(u + v)/2$; with $v = u + at$, average velocity = $(u + u + at)/2 = u + (1/2)at$. Displacement = average velocity $\times$ time = $[u + (1/2)at] \times t = ut + (1/2)at^2$. (b) Tabulate using $s = 5t + t^2$: $t = 0 \rightarrow s = 0$; $t = 1 \rightarrow s = 6 \text{ m}$; $t = 2 \rightarrow s = 14 \text{ m}$; $t = 3 \rightarrow s = 24 \text{ m}$; $t = 4 \rightarrow s = 36 \text{ m}$.

Standard derivation followed by a numerical table.

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Q10. (a) Sketch the v - t graph of a body that is dropped from rest, takes 3 s to fall, hits the ground and bounces back with no energy loss, then rises for 3 s before being caught at the original release height (take downward as positive, $g = 10 \text{ m/s}^2$). (b) Find the total distance and total displacement covered.

Answer: (a) Phase 1 (0–3 s, falling): straight line from (0, 0) to (3, 30) with slope +10. At $t = 3 \text{ s}$, instantaneous bounce: v jumps from +30 to -30. Phase 2 (3–6 s, rising): straight line from (3, -30) to (6, 0) with slope +10. The graph has two segments meeting at the bounce, forming a 'V' shape mirrored about the t -axis. (b) Total distance: fall ($h = \frac{1}{2}gt^2 = 45 \text{ m}$) + rise (45 m) = 90 m. Total displacement: starts at $h = 0$, ends at $h = 0 \rightarrow$ displacement = 0.

Free-fall kinematics + symmetry; distance and displacement diverge here.

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